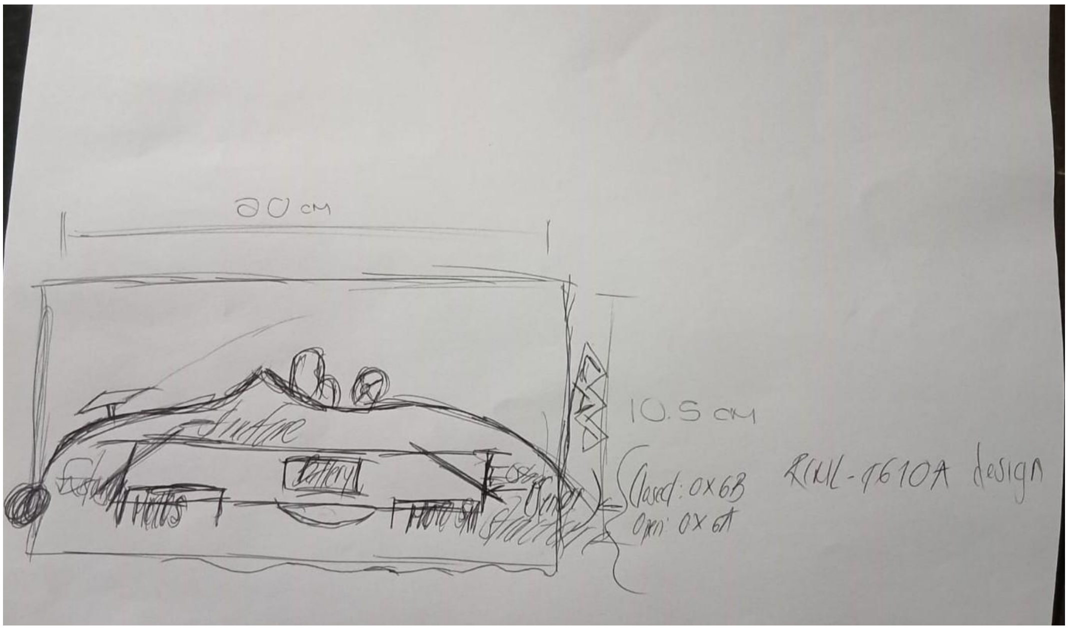
Tuka Curious Minds
MASTER ROBO WARS DESIGN CONCEPT SUBMISSION

Robot Name: TURBO Ceros-T
Weight Classification: 5.kg Max Limit
Footprint Matrix: 420mm (Width) x 420mm (Length) x 55mm (Sleek Casing Height)

Inspiration Concepts discussed, sketched and AI generated to get to try generate a CAD design.

*Note the team are new to CAD, was a steep learning curve but tried to navigate through AI assistance, senior TCM members and YouTube tutorials

Version #1
Scrapped not feasible



We were thinking of doing a race car initially and reconsidered the material and components we had. We then thought there are 3 girsl and 1 boy on the team with male coaches also and changed our mind to try out Robo Wars as mostly girls in this male dominated competition and career right?

Let’s do this! Let’s GO!

Game Play 

Max damage to opponents??

Defence

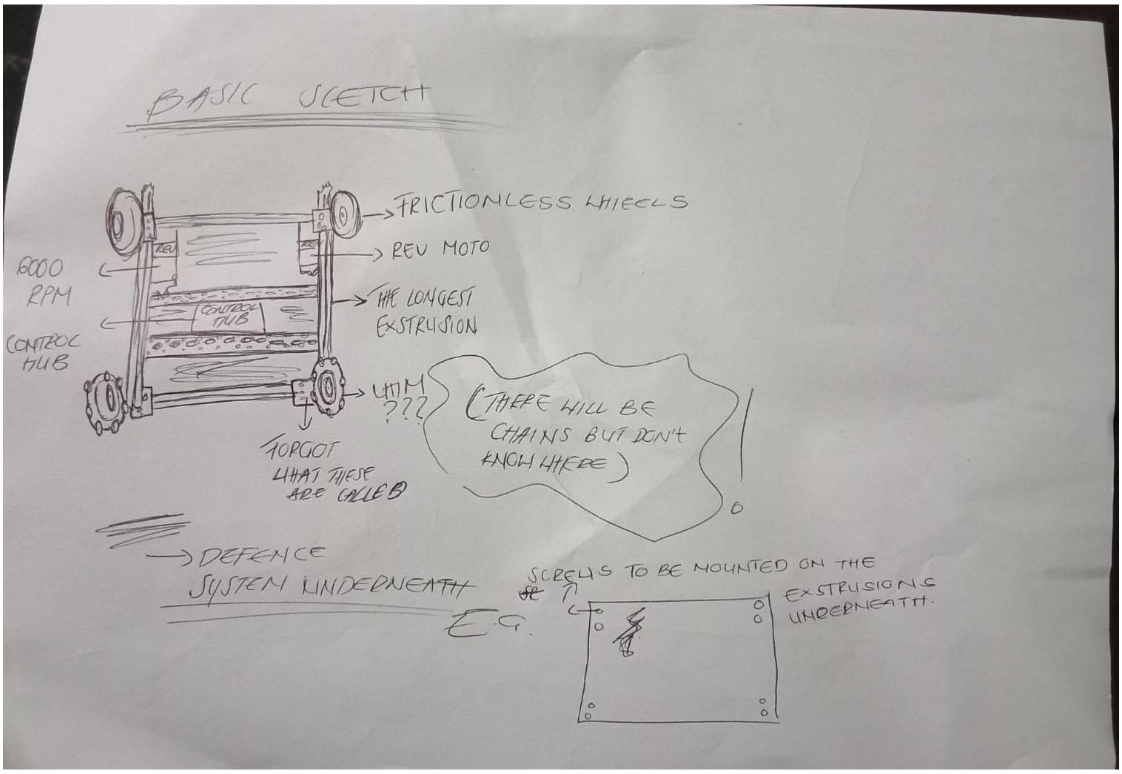
Speed – [running away!]

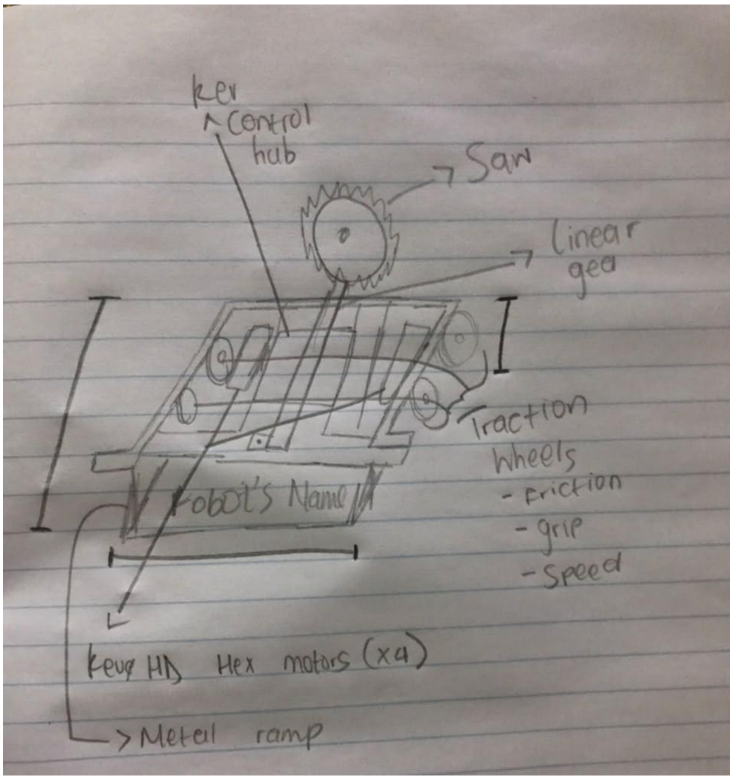
Last one standing…

Version 2

To consider – component parts such as traction wheels, extrusions vs c-channel, available motors.

Weapon system – 2 – 3 blades mounted on front of robot???





<u>Version 3</u>

Material - GoBuilder /Rev

Components. C channels that will hold onto the four traction wheels, because of their grip, speed and friction and then two on each side.

Motors - Single motor that drives both the wheels on each side.

- REV HTX motor, 6000 RPM, or
- GoBuilder 1:1 Yellow Jacket

Chain – Join that using a chain

Control - REV control hub

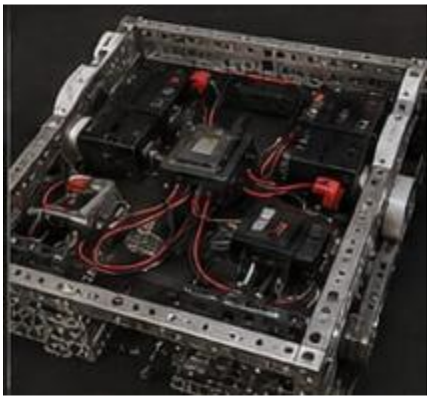
Weapons - solenoid that will stick out of the robot using a linear gear.

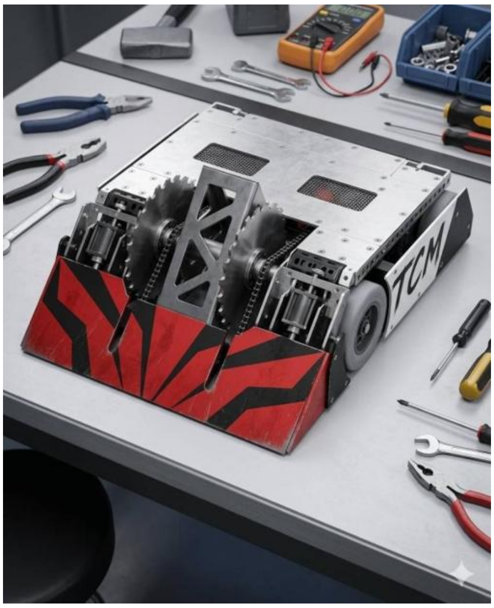
Front - A metal ramp that can pull, push, or lift up the opponent's robot and flip it over. motor, 6000 RPM, and we must make sure each motor has a good torque + speed for wheels

FINAL DESIGN THOUGHTS…………

FINAL CHOICE DESIGN

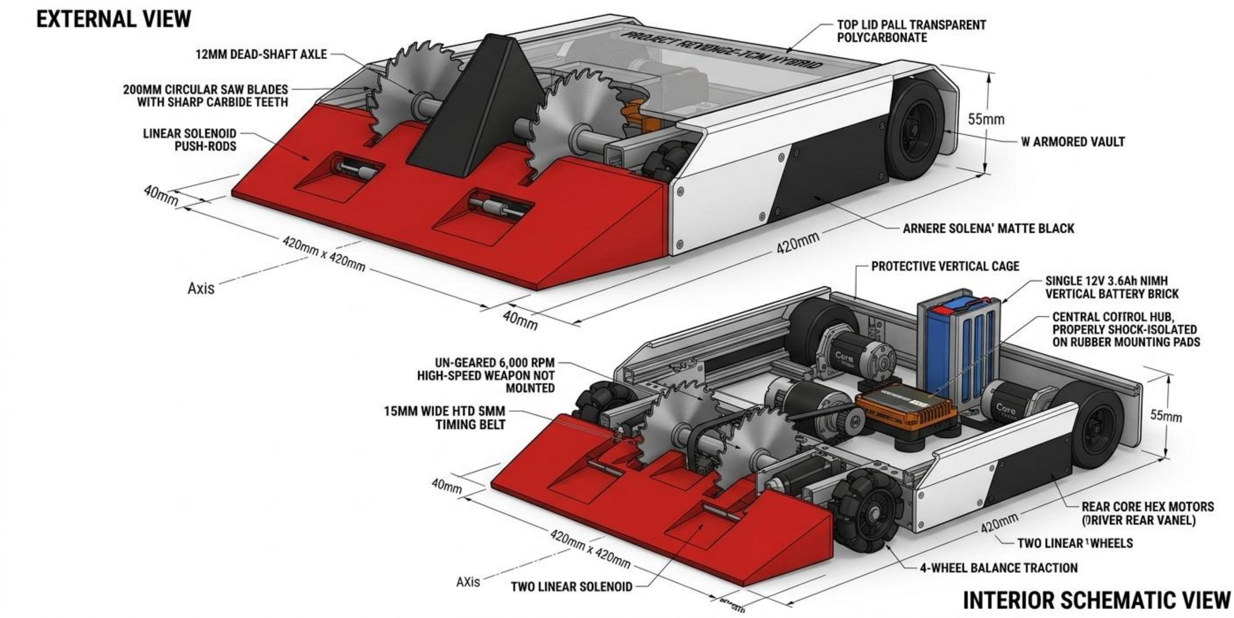
So we took all discussions and sketches and ran through AI to get a feel of what it could look like and compared all possible designs.





MEET TURBO CEROS-T!

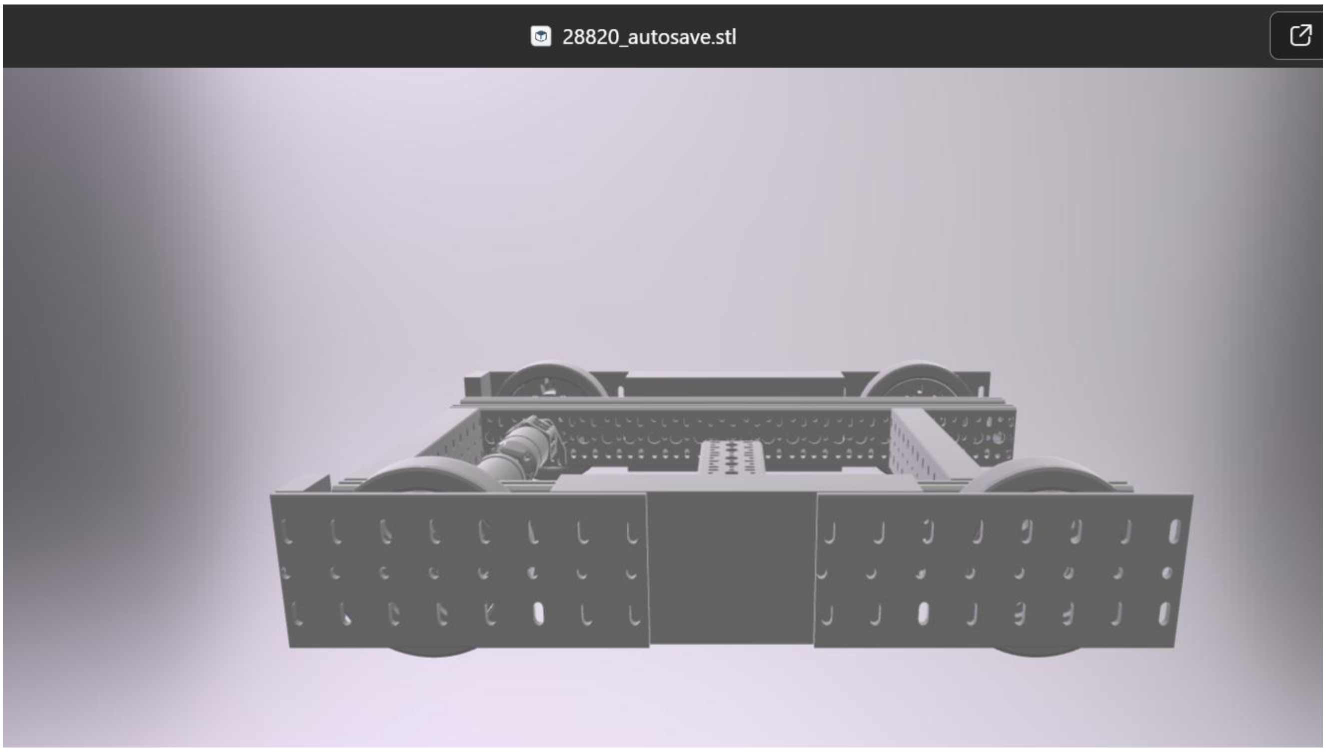
Well what she'll possibly look like 😁



Final iteration [AI generated not 100% placement of blades, and components but this is the concept

Now for Blender/CAD attempt

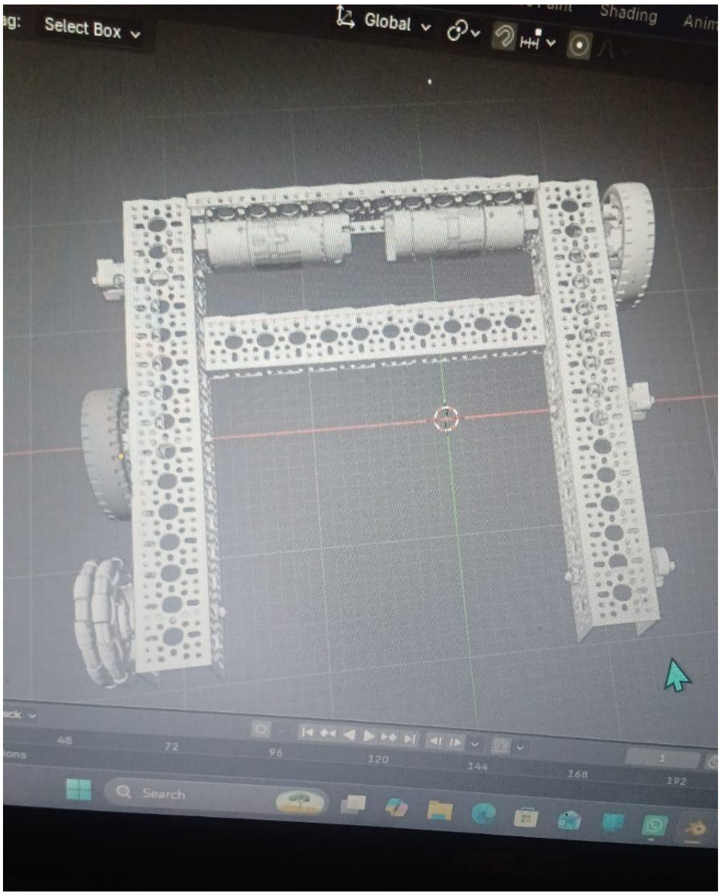
CAD #1

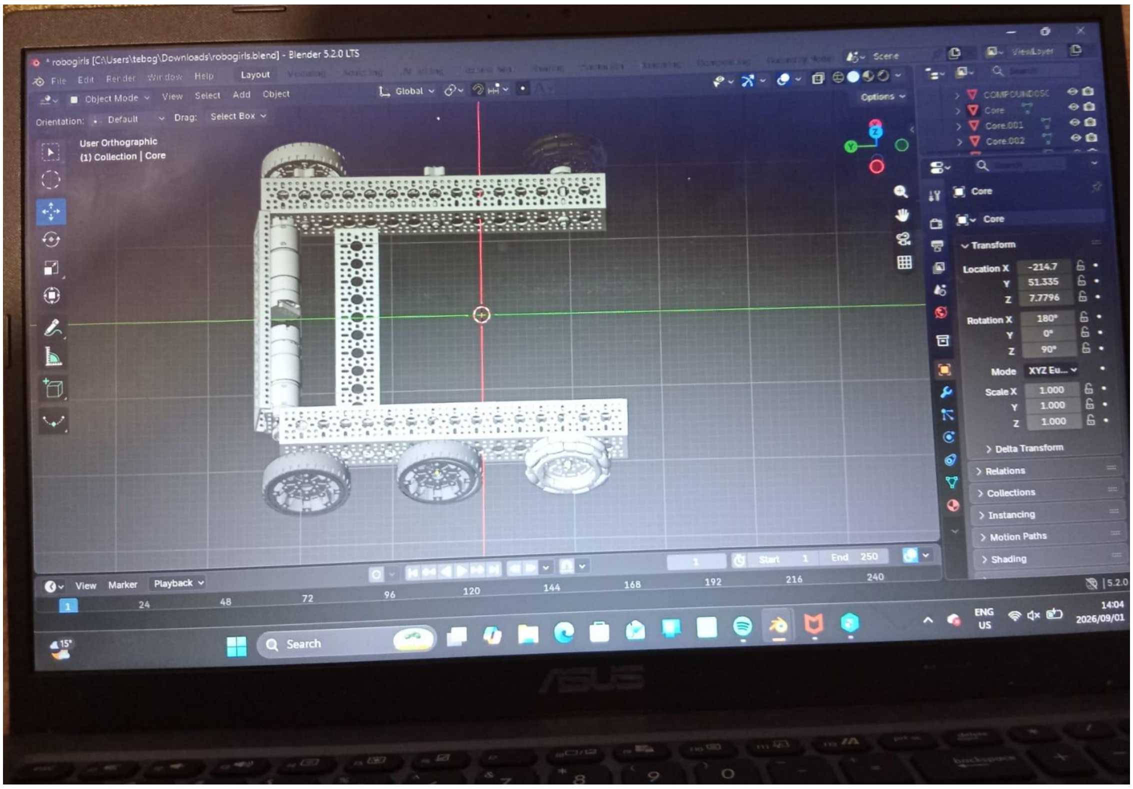


Thank you to senior coach Bonolo for getting us started and coaching session, but this is not easy as he makes it look!

We tried our best….

CAD #2





Not sure if the problem is the laptop, but the ASUS we using keeps stalling, crashing, restarting. This is how far we got

Should we get selected for next round, we hope to get a chance and opportunity to learn the basics and fundamentals of CAD as this was our first attempt as the girls squad trying to understand to do this.

Thank you for considering TURBO Ceros-T to become one of the dreams making it through to the finals to compete against the boys!